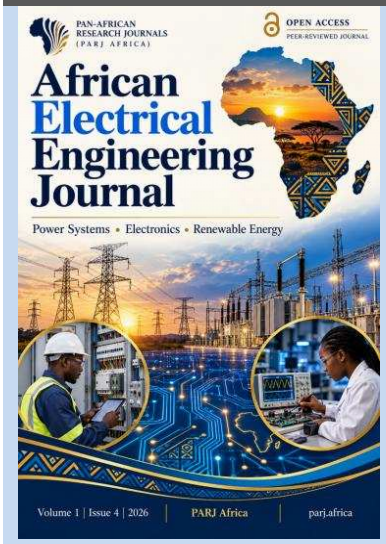
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DESIGN AND IMPLEMENTATION OF A SMART WATER METERING SYSTEM WITH GSM FOR RESIDENTIAL USE

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ABSTRACT

Background: The component cost was GH₵980, equivalent to approximately US$84.12 at the exchange rate reported during the project. Objective: This study designed and implemented a GSM-enabled smart water metering system that integrates an ESP32 microcontroller, YF-S201C flow sensor, MPU6050 gyroscope, SIM800L GSM module, solenoid valve, liquid-crystal display, micro-SD storage, buzzer, and Blynk cloud connectivity. Methods: Effective residential water management requires accurate metering, timely access to consumption data, and rapid detection of tampering. The prototype was designed, simulated, assembled, programmed in the Arduino IDE, and tested under system initialisation, closed- and open-valve water-flow conditions, remote monitoring and control, tamper detection, and graphical data-visualisation scenarios. The implemented system successfully measured water volume in real time, displayed and stored consumption data, enabled remote valve control, transmitted cloud data, and generated SMS and audible alerts when tampering was detected. Results: The prototype therefore demonstrates a practical low-cost approach for improving residential billing accuracy, water-use monitoring, operational efficiency, and revenue protection. Conclusion:

Keywords: *smart water metering; GSM; ESP32; Internet of Things; tamper detection; Blynk; residential water management*

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1. Introduction

1.1 Background of the Study

The increasing water demand is making efficient water distribution crucial in enhancing the quality of life (Gravande *et al*., 2020). Utility water companies supply water to most rural and urban areas, relying extensively on water meters to read and bill customers. Water meters measure the volume of water usage in residential, commercial, and industrial settings, generating bills for the amount consumed and helping to reduce water wastage.

Similar to the majority of nations, Ghana primarily uses analogue water meters, which necessitate manual readings and are frequently inaccurate. Additionally, reading these meters by hand is labour-intensive, time-consuming, does not provide real-time data access, does not detect leaks, results in inaccurate billing, and is prone to errors, highlighting the need for automation (Gungor *et al*., 2011).

Therefore, a smart water metering system with GSM technology offers a promising solution by enabling real-time monitoring and leak detection, reducing revenue losses, and ensuring efficient water management.

1.2 Problem Statement

Accurate water billing, water conservation promotion, better utility administration and efficiency, and leak detection to minimise water loss all depend on the water metering system. The current water metering system is analogue, which is prone to error and limited data collection, lacks real-time monitoring, has difficulties in leak detection and tampering, which limits customer engagement, requires frequent calibration, and has limited remote monitoring and control, resulting in inaccurate measurement of water usage, revenue losses, and customer dissatisfaction.

This research aims to design and implement a smart water meter that utilises GSM technology for residential use, providing real-time access to water usage data and enabling remote control. The design and implementation of the Smart Water Metering system with GSM for residential use will reduce operational costs, improve efficient water usage management, and avoid losses.

1.3 Study Objectives

The objectives for this project work are to:

Design a Smart Water Metering System with real-time monitoring, control, and a tampering detection system.

Improve overall water management efficiency; and
Reduce operational expenses.

1.4 Study Approach

The methods used for this research are to:

Review relevant literature;
Design a circuit for the proposed system, and build the Prototype.
Program the microcontroller with Arduino IDE; and
Test the metering system design.

1.5 Development Environment and Facilities

The facilities used for the project include the following:
UMaT-SRID Library and internet facilities;
Personal Laptop Computer with Proteus version 8.17 professional and Arduino IDE 2.3.4 software installed; and
UMaT-SRID Electrical workshop.

2. Literature Review

2.1 Overview

This section reviews the water-management context, water-metering technologies, major components used in smart metering, and related studies relevant to the design and implementation of a GSM-enabled residential smart water meter.

2.2 Water Management

Water management is vital for planning, developing, and distributing water resources sustainably and equitably. It involves balancing human use, agriculture, industry, and ecosystem health while addressing issues such as water scarcity, pollution, and climate change. Although water covers 75 percent of the Earth's surface, only 3 percent is freshwater, and an even smaller part is readily accessible. Studies indicate that people require at least 20 to 40 litres of water daily for drinking and hygiene, yet over a billion lack access to clean water (Hemant, 2019).

In conservative agriculture, effective water management practices optimise water use, reduce soil erosion, and enhance soil health. Strategies like smart water metering systems help improve efficiencies in water management and minimise water waste. In contrast, challenges such as mismanagement, population growth, climate variability, and ageing infrastructure must be addressed to ensure efficient water usage (Jat *et al*., 2023). The following sections will discuss water management techniques for efficient and responsible usage.

2.2.1 Smart Water Management System

Smart water management aims to optimise water usage at the regional or city level, focusing on sustainability and self-sufficiency. It utilises innovative technologies, such as IoT sensors, smart pipes, and data analytics, to help reduce leakage, improve water quality, enhance customer experiences, and increase operational efficiency.

2.2.2 Smart Water Metering System

A smart meter measures and transmits consumption data regularly. To manage water resources effectively, sensors or actuators are needed to monitor usage patterns. Unlike traditional meters, which are read monthly, smart meters enable remote readings, providing customers and management with instant access to data. An Advanced Metering Infrastructure (AMI) is essential for handling this information.

2.2.3 Leak Detection System

Water leakage detection systems utilise acoustic and ultrasonic sensors, along with software algorithms, to identify leaks and anomalies in water pipelines. Leak detection methods include measuring the difference between fluid inflow and outflow, as well as analysing Negative Pressure Waves (NPWs). A significant discrepancy detected by differential pressure sensors along the pipes can signal a leak.

2.2.4 Rainwater Harvesting System

Rainwater harvesting involves collecting and storing rainwater from surfaces like rooftops and terraces for on-site use. This water can be utilised for toilet flushing, laundry, garden irrigation, and car washing, all aimed at

reducing reliance on centrally supplied drinking water. These household activities account for approximately 80 - 90 % of global water consumption, underscoring the conservation benefits of rainwater harvesting.

2.3 Water Metering Systems

An important water supply network management device is the water meter, which measures the volume of water supplied from a public water network distribution system to a residential or commercial setting, providing water usage data to utility companies to help generate bills for customers (Davidson *et al.,* 2006). Below are the discussions on some water meter types.

2.3.1 Mechanical Meters

Mechanical water meters are effective devices that utilise mechanical components to measure water flow accurately. Many countries choose fully mechanical water meters due to their affordability and dependability. While these meters require manual readings, their reliability makes them a popular choice. Water flow measurement can be based on either velocity or displacement, with common types including impeller meters and turbine meters, which feature designs such as oscillating pistons and nutating discs.

Velocity measurement meters

Velocity measurement meters, specifically jet meters, operate by harnessing the power of single or multiple jets striking a radial-vaned impeller within the meter. Figure 2.1 (Anon., 2019) shows images of jet meters.

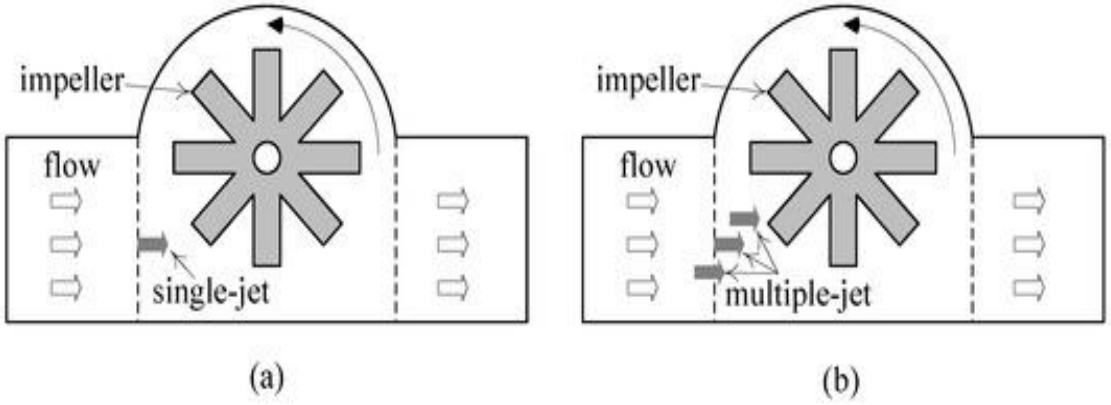


Figure 2.1 (a) Single Jet Meter, (b) Multiple Jet Meter

Positive displacement meters

Positive displacement meters, such as oscillating piston and nutating disk meters, measure water volume by dividing it into discrete quantities. Oscillating piston meters count the fills and empties of a chamber with a rotating piston. Nutating disk meters use a nutating disc to achieve similar measurements.

2.3.2 Electromechanical Water Meter

In the last twenty years, the integration of electronic circuit components into mechanical water meters has revolutionised their functionality, paving the way for advanced features such as Automatic Meter Reading (AMR) (Wu *et al.,* 2023). This innovation not only enhances accuracy but also streamlines the monitoring process, making it more efficient for both consumers and utility providers.

2.3.3 Electronic Water Meter

Electronic water meters, including electromagnetic, fluidic, and ultrasonic types, are more accurate than mechanical meters, enhancing water management in smart cities (Garmabdari *et al*., 2012). They provide precise monitoring and leak detection. By integrating these electronic meters with real-time sensors, utilities can reduce

water loss and optimise billing, promoting water security and sustainability. Figure 2.2 shows an electromagnetic and a fluidic meter (Li and Chong, 2019).

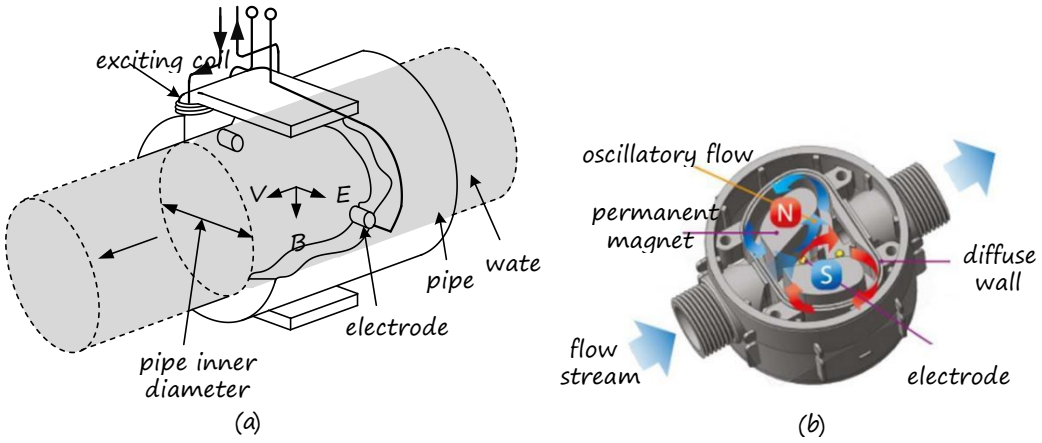


Figure 2.2 (a) Electromagnetic Meter, (b) Fluidic Meter

2.3.4 Smart Water Meter

Smart-meter implementation is motivated by several factors, including reducing commercial losses, meeting legal requirements, implementing climate policies, addressing capacity shortages, and facilitating demand-side management (Kinuthia, 2019). Smart meters facilitate advanced data collection, including tamper detection, flow logging, and temperature monitoring. Although ultrasonic meters come with a higher initial cost, they provide long-term savings due to their improved accuracy, less frequent maintenance (with battery replacement needed only every 15 - 20 years), and superior data integration compared to mechanical meters (Beatty, 2016). Their durability surpasses that of traditional meters, making them a cost-effective choice for modern water management. The diagram in Figure 2.3 shows a typical smart water meter setup (Anon, 2018a).

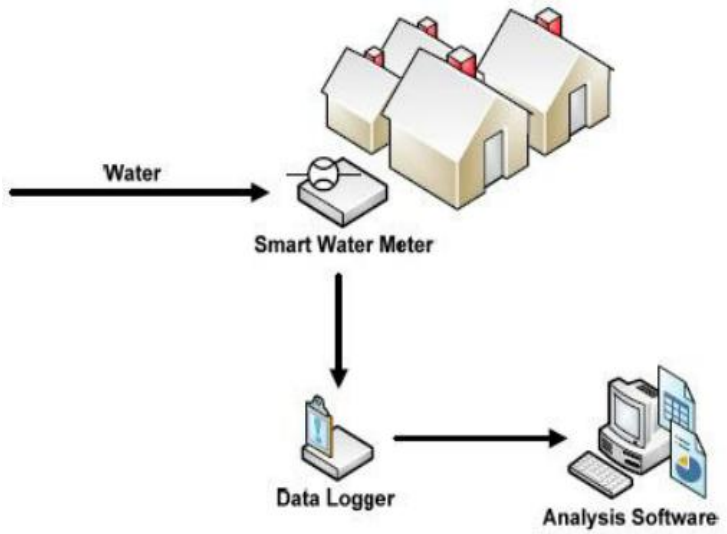


Figure 2.3 Typical Smart Water Meter Setup

2.4 Components Used for Smart Water Metering System

Components such as a microcontroller, GSM module, solenoid valve, water flow sensor, and acoustic sensor work together to provide a comprehensive smart water metering system that enables real-time monitoring, leak detection, automated billing, and data-driven decision-making.

2.4.1 Microcontroller

A microcontroller (MC) or microcontroller unit (MCU) is a compact computer built on a single chip. It contains one or more Central Processing Units (CPUs), memory, and programmable input/output components. The chip typically includes program memory, which can be NOR flash, one-time programmable RAM (OTP RAM), ferroelectric RAM, and a small amount of RAM for temporary data storage. Microcontrollers are designed for embedded applications, meaning they are specialised for specific tasks, unlike microprocessors used in personal computers or general-purpose devices, which often use separate chips. Figure 2.4 (Anon., 2018b) shows a block diagram of a microcontroller with supported devices.

2.4.2 GSM Module

The GSM (Global System for Mobile Communications) module is a vital component of mobile communication technology, remaining relevant alongside newer standards such as 4G and 5G. Its key components are the Mobile Station (MS), Base Station Subsystem (BSS), Network Switching Subsystem (NSS), and Operation Support System (OSS), which ensure reliable voice and data transmission. GSM offers global coverage and compatibility with older systems, making it a cost-effective option for basic communication. It is also essential for smart water metering, facilitating remote data transmission, automated meter reading, leak detection, tampering alerts, and customer notifications, all of which enhance accuracy and reduce operational costs. Figure 2.6 (Sukhno and Sotnik, 2024) illustrates these main components.

2.4.3 Water Flow Sensor

The water flow sensor measures the amount of water transferred from one area to another. It comprises a plastic valve body, a water rotor, and a Hall effect sensor. These sensors measure the volume of water flown to each house (Anandhavalli *et al.,* 2018).

2.4.4 Solenoid Valve

Solenoid valves are electro-mechanical devices that control water flow in metering systems by converting electrical signals into mechanical motion. They provide automated shut-offs for leaks, enable remote flow control, and support demand-based distribution, improving efficiency and reducing water waste. With fast response times and IoT compatibility, Solenoid valves are ideal for irrigation, residential meters, and industrial applications. Regular maintenance of coils and seals is crucial for optimal performance. Figure 2.5 (Anon, 2024) shows a three-way solenoid valve diagram.

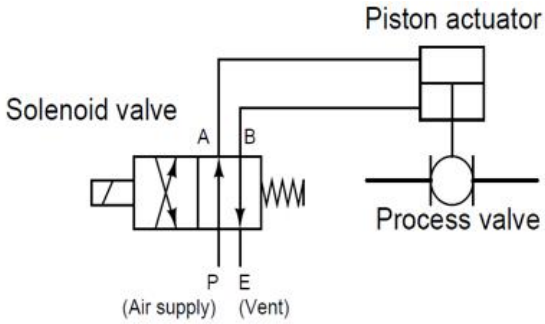


Figure 2.5 Three-Way Solenoid Valve Diagram

2.5 Review of Related Works

The review of related works on smart water metering systems is summarised in the tables below. It includes the author’s name, title, methods or components used and each work's limitations (s).

Table 2.1 Review of Related Works on Smart Water Metering System

SN	Author	Title	Methods/ Components Used	Results	Limitations
1.	Gajanan et al., (2017).	Smart Water Flow Monitoring and Forecasting System.	Raspberry Pi, Arduino, Hall Effect sensor-based flow meter, Cloud infrastructure, Database, Web-based solution.	The system provided real-time water usage monitoring, accurate measurement, and predictive analytics.	Potential data transmission issues exist, with no information on tampering detection.
2.	ha *et al.* (2018).	Smart Water Monitoring System for Real-Time Water Quality and Usage Monitoring.	Smart Water Quantity Meter, Smart Water Quality Meter, cloud-based data storage, SMS notification system.	The system continuously monitors water quality and usage in real-time.	Scalability challenges and dependence on consumer cooperation.
3.	Amir *et al.,* (2022).	Smart Water Meter for Automatic Meter Reading.	The system has a main valve, a flow meter sensor, and a node MCU (microcontroller) with a wireless transceiver.	The flow meter sensor operated accurately with a maximum error of 0.3 litres.	Lack of security and privacy of the data transmitted to the web server.
4.	Rahman et al., (2018).	Automated Household Water Supply Monitoring & Billing System.	Arduino Mega 2560, Double Relay, Water Level Sensors, Water Flow Sensors, DC Water Motor.	The system accurately measures water usage, displays usage in real-time, and generates automated bills based on usage.	Vulnerability to power outages.
5.	Satiya *et al.,* (2017).	Optimisation of Water Consumption Using a Dynamic Quota-based Smart Water Management System.	Local computing, Human Machine Interface (HMI), IoT-enabled mobile-friendly web interface, GSM module, Machine learning algorithms.	The system offers optimised water consumption, reduced losses, and real-time monitoring.	It requires significant infrastructure investments.

Table 2.1 cont'd

SN	Author	Title	Methods/ Components Used	Results	Limitations
6.	Megantoro *et al.,* (2020).	Validation Method for Digital Flow Meter for Fuel Vendor.	Arduino Mega 2560: Microcontroller, Positive Displacement Flow Meter, Solenoid Valve, 4×20 LCD, and Keypad: The software used is Arduino/C programming for volume calculation, valve control, and LCD output.	It successfully measures an average error of 2.24% (within acceptable limits for retail applications). Volume-Specific Errors: 500 mL: +14 mL (2.8% error). 2000 mL: -46 mL (2.3 % error).	There is no discussion about alternative power supplies during a power outage.
7.	Randall *et al.,* (2019).	Smart Water Metering Technology for Water Management in Urban Areas.	Smart water meters in the Port Macquarie-Hastings Council (PMHC) use Microsoft Excel and IBM SPSS to analyse consumption trends and variability.	It showed that water consumption is highest in summer and lowest in winter.	There is no discussion on alternative power supply when there is a power outage.
8.	Tan *et al.*, (2018).	Automatic Power Meter Reading System Using GSM Network.	A single-phase digital kWh power meter compliant with IEC61036 standards.	The system successfully demonstrated automatic meter reading, billing, and notification.	There was no real-world testing of the prototype.
9.	Martinel *et al*., (2023).	Water Meter Reading for Smart Grid Monitoring.	The proposed method uses YOLOv5, a deep learning model, for digit localisation and dial-meter recognition.	The YOLOv5 model successfully localises and recognises digits in the dial meters, showcasing its potential for automating meter readings.	Lacked accurate readings.

Table 2.1 cont'd

SN	Author	Title	Methods/ Components Used	Results	Limitations
10.	Sarangle *et al.,* (2022).	GSM-Based Power Meter Reading and Control System.	GSM Modem, Microcontrollers, Power Meter, Triac, and Zero-Crossing Detector, SPI (Serial Peripheral Interface).	The system successfully reads power consumption data from the meter and transmits it to the electricity provider and the user via SMS.	The system does not offer real-time monitoring.
11.	Rajput *et al.,* (2018).	Modelling of Arduino-based Prepaid Energy Meter using GSM Technology.	Arduino UNO, GSM Module, Energy Meter, Relay and Relay Driver, LCD Display, Opto-coupler**.**	The system sends SMS notifications to users regarding their energy balance, recharging status, and load disconnection due to insufficient balance.	There was no discussion on theft detection.
12.	Mir *et al.,* (2019).	Review of Smart Electric Metering System Based on GSM/IoT.	Arduino microcontroller, GSM modem, Relay circuit, Energy meter, IoT integration.	The system archived Real-time monitoring, prepaid billing, and tampering detection are cost-effective.	There is no discussion about alternative power supplies during a power outage.
13	Pratibha *et al.,* (2019).	Smart Water Monitoring System for Real-Time Water Quality and Usage Monitoring.	Flow sensor, GSM,pH sensor, Conductivity sensor, Temperature sensor,Oxidation Reduction Potential (ORP) sensor, Microcontroller.	The system achieved the following results: Real-Time Monitoring, Automated Alerts and Reduced Human Error.	There were no explicit details on power management and no discussion on leak detection.

Table 2.1 cont'd

SN	Author	Title	Methods/ Components Used	Results	Limitations
14.	Li *et al.,* (2019).	Design and Implementation of a Self-Powered Smart Water Meter.	It uses a micro-hydro turbine to generate electricity. It has adjustable turbine dimensions for different pipe sizes, and calibration via remote algorithm updates. Signal Processing: Rectified voltage measurements with digital filtering to estimate flow rates.	Accurately measures flows between 200 and 650 L/h. Harvested energy powers the meter, with peak consumption at 137 mW. Charges batteries to power the meter, eliminating the need for external power.	Low Flow (< 200 L/h): Turbine fails to spin, limiting leak detection. High Flow (> 650 L/h): Turbine slip and magnet saturation reduce accuracy.
15	Palau *et al.,* (2019).	Numerical Study of Upstream Disturbances on the Performance of Electromagnetic and Ultrasonic Flow Meters.	Electromagnetic, ultrasonic flow meters, Gate valve, butterfly valve, Fluent 6.1 with Gambit, RANS approach, and Comparison of the experimental data.	Overestimation and underestimation in the flow turbulence, High sensitivity in the butterfly valve, and only the butterfly valve at 3D/6D from the EM flowmeter met the ±2 % accuracy standard.	The system required a distance greater than 6D for acceptable accuracy, limiting its practical use in tight spaces, and severe distortions occurred at greater than 50% closure, making error correction difficult. It was limited to 80 mm pipes.

2.6 Summary of Related Works

Implementing a smart water metering system is essential for effective water management. This approach ensures accurate measurement of water flow, leading to precise billing and improved efficiency compared to traditional methods. While many studies emphasise leak detection, they often overlook practical solutions and the need for a reliable backup plan for power management. Addressing these gaps is crucial. Our initiative aims to tackle these challenges and enhance water management practices.

3. Methodology

3.1 Methodological Overview

Water wastage caused by undetected leaks, inadequate monitoring, and inefficient management remains a significant challenge in domestic and industrial settings. To address this problem, a smart water metering system was developed to monitor water usage automatically, detect tampering, control the water supply when required, and alert users in real time through SMS. The prototype uses an ESP32 microcontroller integrated with sensing, storage, communication, display, and actuation modules to provide an autonomous and cost-conscious water-management solution.

3.2 System Design

The design system describes the design concept and the design criteria.

3.2.1 Design Concept

Figure 3.1 illustrates the block diagram of the proposed Smart Water Metering System with GSM for residential applications.

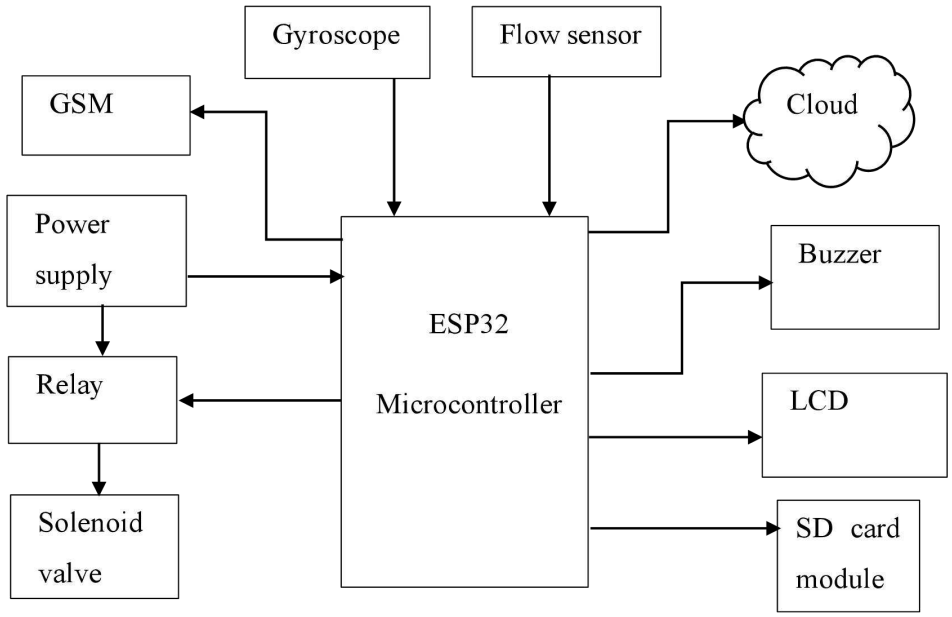


Figure 3.1 Block Diagram of the Proposed System

The proposed smart water metering system is centred around an ESP32 microcontroller that manages data from sensors, communicates with external platforms, and controls system responses. A flow sensor connected to the ESP32 measures water consumption, while a gyroscope detects tampering by sensing abnormal orientation changes. The microcontroller receives power from a regulated power supply and

displays real-time readings on an LCD screen. For immediate alerting, a buzzer is triggered in the event of tampering, and an SD card module stores consumption logs locally for offline access. A relay controls a solenoid valve to automatically shut off water in the event of leaks or tampering. The GSM module (SIM800L) sends SMS alerts specifically during instances of tampering. Additionally, the ESP32 utilises its Wi-Fi module to upload water usage data to a cloud platform every six hours, allowing for remote monitoring and graphical analysis of consumption patterns. The cloud also allows remote control of the meter, making the system suitable for automated, secure, and efficient water management.

3.2.2 Design Criteria

The design criteria outline the official objectives that the proposed system must satisfy during its implementation. Below are the key criteria for the system:

The system should provide a mechanism to enable real-time monitoring and tampering detection;
It should enable consumers to remotely manage their water consumption to improve overall water management efficiency;
The system should help utility providers to reduce operational expenses; and
The proposed system should provide a database to the water management team to have easy access to records when needed.

3.3 System Operation

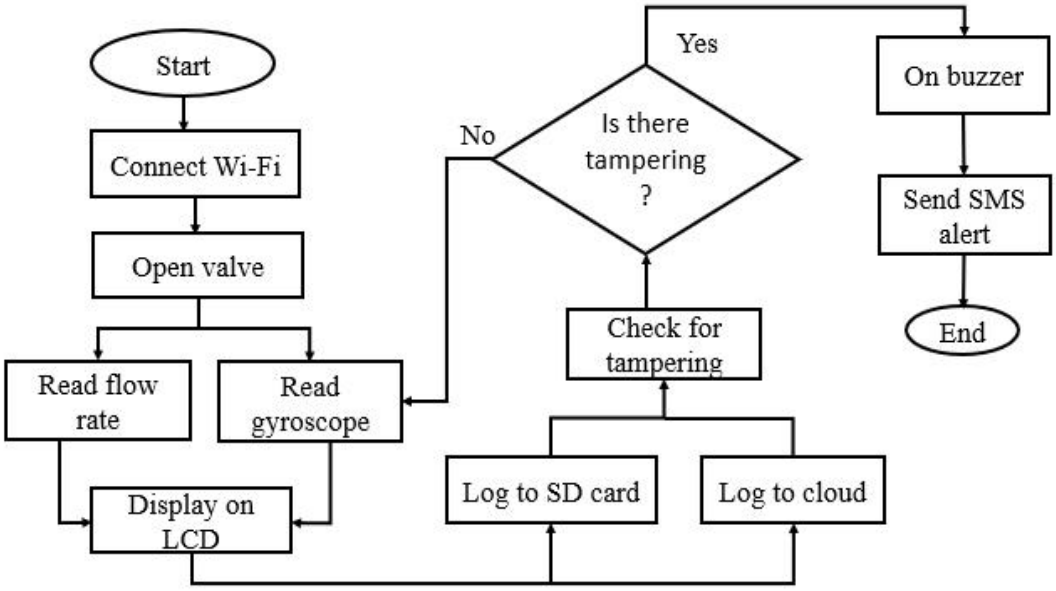


Figure 3.2 Flowchart of the System Operation

The flowchart begins with the system starting and immediately establishing a Wi-Fi connection to enable network-based operations. Once connected, the valve is opened to allow water flow through the system. The system then performs two key readings simultaneously: it measures the flow rate and reads data from the gyroscope. The measured flow rate is subsequently displayed on the LCD for monitoring. After these readings, the system checks for any tampering that may have occurred. If no tampering is detected, the process loops back to continue reading the flow rate and gyroscope data in a continuous cycle. However, if tampering is detected, the system proceeds to check for tampering details, logs the event data to both an SD card and the cloud for storage, activates the buzzer to signal an alert, sends an SMS notification about the tampering incident, and then concludes the process.

3.4 Specification and Selection of Components

This section outlines the implementation of the Smart water metering System with GSM, showing how to convert the theoretical design into a functional prototype. It includes assembling hardware components, integrating sensor modules with the ESP32 microcontroller, and developing embedded software. The design ensures flexibility and scalability, enabling the systematic development, testing, and integration of individual subsystems within the system. Table 3.1 presents a summary of the components used in implementing the proposed system and their various applications.

Table 3.1 List of components and their functions

SN	Component	Function
1.	**ESP32 Microcontroller**	**The microcontroller executes tasks such as reading inputs, processing data, and controlling outputs.**
2.	Active Buzzer	**Triggers a sound when an electric current flows through it.**
3.	**Flow sensor (YF-S201)**	**Measures the flow rate of water passing through it.**
4.	**Gyroscope**	**Measures the orientation and tilt of the meter.**
5.	**12v Solenoid Valve**	**Opens to allow the flow of water when energised.**
6.	**GSM SIM800L Module**	**Sends SMS messages or data to report water usage and alerts.**
7.	**DC-DC Buck converter**	**Convert the 12 V DC voltage to a 5 V DC voltage.**
8.	**16x4 LCD**	**Provides a visual interface for users to view meter status.**
9.	**Micro SD card module**	**Provide local storage capacity for data logging.**
10.	**Lithium battery**	**Provides a stable power supply.**

3.4.1 Espressif System Processor 32 Microcontroller (ESP32)

The ESP32 microprocessors are cost-effective, energy-efficient SoC devices featuring integrated Wi-Fi and Bluetooth, suitable for portable and IoT applications. They operate reliably from -40 °C to +125 °C and include dynamic power management, cryptographic hardware acceleration, and a low-power co-processor. With a 32-bit RISC architecture running at 160 MHz and dual cores, the ESP32 supports 802.11 b/g/n Wi-Fi and Bluetooth 4.0 LE. It requires a 5V power supply and has 38 general-purpose I/O pins with support for various protocols. The ESP32-WROOM-32 is ideal for smart water metering due to its robust design, wireless communication capabilities, and multi-protocol support for real-time monitoring and reporting. The ESP32 microcontroller's picture and pin layout are shown in Figures 3.4 (Anon., 2023b) and 3.5 (Anon., 2023c), respectively.

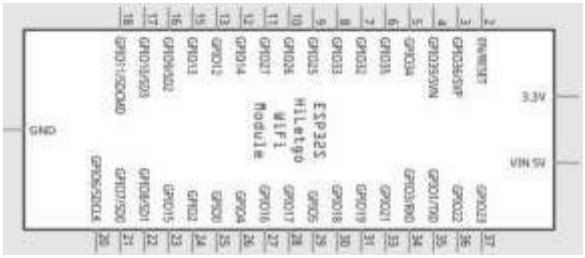


Figure 3.3 A Pin Layout of ESP32 Microcontroller

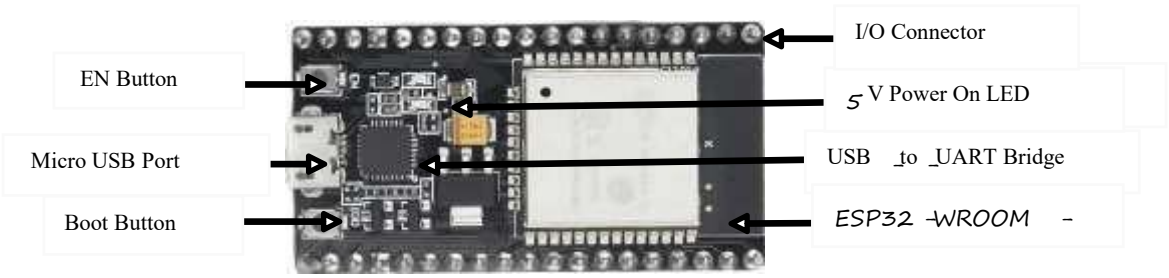


Figure 3.4 An ESP32 Microcontroller

3.4.2 Sensor Modules

Sensor modules measure water flow through pulses and detect leakages or anomalies, converting them into electrical signals that the microcontroller analyses to make decisions in smart water metering systems. Additionally, a gyroscope is used for detecting tampering.

Water Flow Sensor

The Flow Sensor (YF-S201 is a Hall-effect-based flow sensor suitable for water metering. It operates at 5 - 18 V DC and can measure flow rates in the range of 1 - 30 L/min. Its built-in impeller and magnetic sensor generate digital pulses proportional to flow, which can be counted to calculate volume. It is selected for its affordability, ease of use, and compatibility with Arduino input pins for accurate water volume monitoring. Figure 3.5 (Anon., 2022) shows a picture of a YF-S201 1-30 L/min Water Flow Hall Effect Sensor.

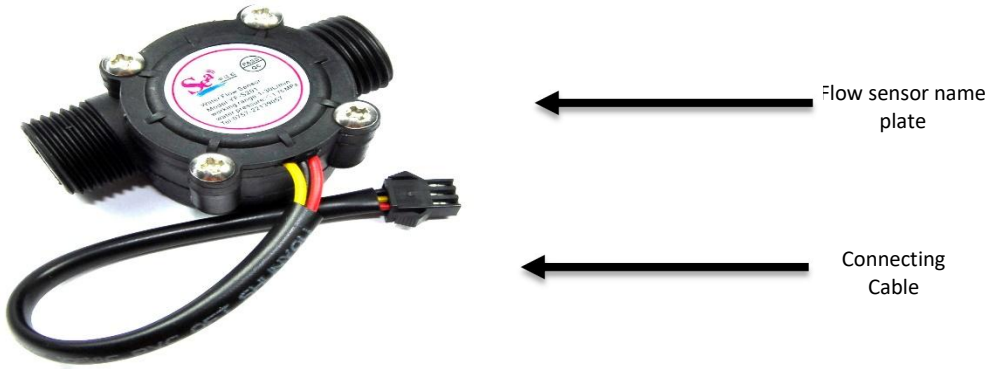


Figure 3.5 YF-S201 Water Flow Hall Effect Sensor

Gyroscope

A gyroscope sensor, like the advanced MPU6050, acts as a vigilant protector against tampering by monitoring angular velocity across three axes: X, Y, and Z. When the device remains still under normal conditions, it effectively detects irregular motions caused by shaking, tilting, or rotating, signalling potential tampering. With a threshold set at ±300 degrees per second, any breach on an axis activates the ESP32, which swiftly takes action. It closes the solenoid valve to prevent interference, sends alerts via the GSM module, displays a warning on the LCD, and logs the incident on the SD card. This comprehensive response enhances security, especially in remote or unattended installations, ensuring robust protection against tampering. Figure 3.6 is the MPU6050 Gyroscope used in this work.

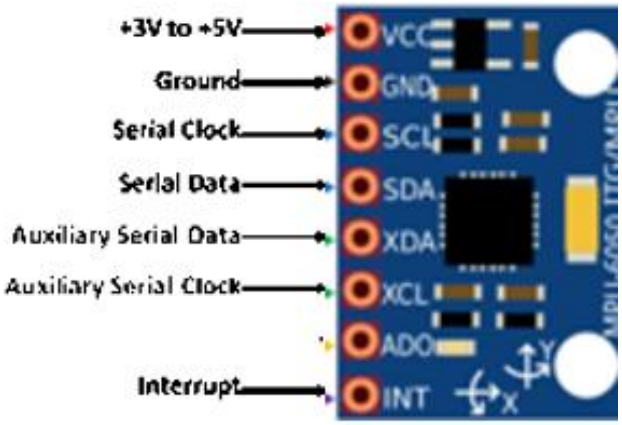


Figure 3.6 MPU6050 Gyroscope

3.4.3 Output Devices and Actuators

Output devices and actuators are components that initiate actions based on signals from the microcontroller. They include relays, solenoid valves, buzzers, and liquid crystal displays (LCDs).

Relay (5 V)

The 5V single-channel relay enables the microcontroller to control high-voltage or high-current devices, such as solenoid valves. It acts as a switch that can be safely triggered by a low-power digital pin from the Arduino. Selection is based on switching capacity, opto-isolation for safety, and response time.



Figure 3.7 5 V Relay

Solenoid Valve (12 V)

The Solenoid valve electromechanically operates to regulate water flow. A typical 12V DC solenoid valve controlled by a relay is used, which opens when voltage is applied. It allows for the remote shutdown of the water supply via a relay triggered by the microcontroller, especially during excessive consumption or when detected leaks occur, and when the flow reaches the specified timestamp. Figure 3.7 (Anon., 2020) shows a 12 V solenoid valve.



Figure 3.8 12 V Solenoid Valve

Buzzer

An active buzzer is a piezoelectric device that produces sound when connected to a DC voltage source. With a sound pressure level of 85 dBA at 10 cm and a current draw of less than 15 mA, the buzzer operates (within a 4 - 8 V range) with a rated current of 30 mA. It emits a continuous tone at 300 Hz and operates

within a temperature range of -25 °C to 80 °C. The smart water metering system provides audible alerts for key events, such as leakage detection and unauthorized tampering. The buzzer is chosen for its low power consumption and reliability, effectively drawing attention to critical issues to prevent water loss.

16x4 I2C LCD

The I2C LCD, measuring 16×4, features an I2C interface to display 16 characters across two lines as shown in Figure 3.9. The smart water metering system displays real-time data, including water flow rate, total consumption, system status, and alerts. Its I2C interface simplifies wiring and reduces the number of microcontroller pins needed. An in-built potentiometer allows for contrast adjustment, and the LCD operates at 5V, making it compatible with various microcontrollers, including the ESP32, ideal for embedded IoT applications.

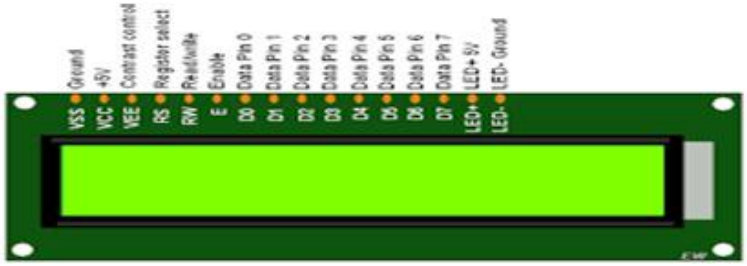


Figure 3.9 16x4 I2C LCD

3.4.4 Communication Modules

Communication modules allow for data exchange among components or devices within a system. Typical examples of these modules are Wi-Fi, Bluetooth, and GSM. They enable microprocessors to connect with users' mobile devices and cloud storage, allowing for remote monitoring and control.

SIM800L GSM Module

The GSM module is vital for wireless communication. The SIM800L and SIM900A are popular due to their quad-band GSM/GPRS capabilities (850/900/1800/1900 MHz), which allow for global connectivity. They support standard AT commands and operate on low voltages (3.4V – 4.4V), making them compatible with microcontrollers. Selection focuses on reliable network support, low power consumption, and ease of integration, with support for UART serial communication. It sends an alert of the consumption at the timestamp, and a notification when a leakage or tampering occurs. Figure 3.10 (Anon., 2021) shows an image of the SIM800L GSM Module.



Figure 3.10 SIM800L GSM Module

Wi-Fi Module (Built-in ESP32)

The ESP32 features a robust, integrated 802.11 b/g/n Wi-Fi module that operates on the 2.4 GHz frequency band, supporting both station and access point modes with data transmission speeds of up to 150 Mbps. This makes it ideal for a smart water metering system, allowing for smooth remote monitoring and control of water usage. With real-time communication to cloud platforms like Blynk, users can easily track consumption trends, receive instant leak notifications, and manage settings through a mobile app. The module employs WPA2-PSK encryption for secure data transfer and maintains reliable connectivity within a range of 30 to 50 meters. The ESP32 also features automatic reconnection with exponential backoff algorithms for efficient recovery from network disruptions. Over-The-Air (OTA) firmware updates simplify maintenance and enhance functionality. Furthermore, its low-power modes reduce energy consumption during idle periods, ensuring the smart water metering system stays responsive and promotes sustainable water management.

3.4.5 Micro-SD Card Module

The micro-SD card module is used for local data logging. It supports the standard FAT32 file system, usually with micro-SD cards up to 32 GB. It connects to microcontrollers via SPI (Serial Peripheral Interface). It's selected for its ability to store large amounts of timestamped usage data, especially helpful for backups during GSM transmission failures or power outages. Figure 3.11 (Anon., 2021) shows an image of the micro-SD card Module.

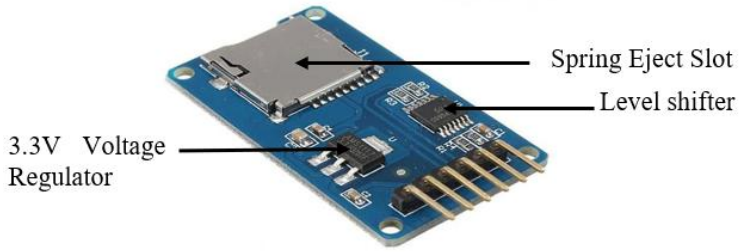


Figure 3.11 Micro-SD Card Module

3.4.6 Cloud Integration Using Blynk

The Blynk platform provides a user-friendly interface that is integrated into the system, enabling real-time monitoring of water usage and control of components connected to the ESP32. The ESP32 microcontroller communicates with the Blynk cloud via Wi-Fi.

3.5 Circuit Design

Figure 3.12 is the K-Card schematic diagram of the design and implementation of a smart water metering system with GSM for residential use. It shows the functional connections between the components of the physical arrangement. It prioritises conveying the logical flow and purpose of the system being designed.

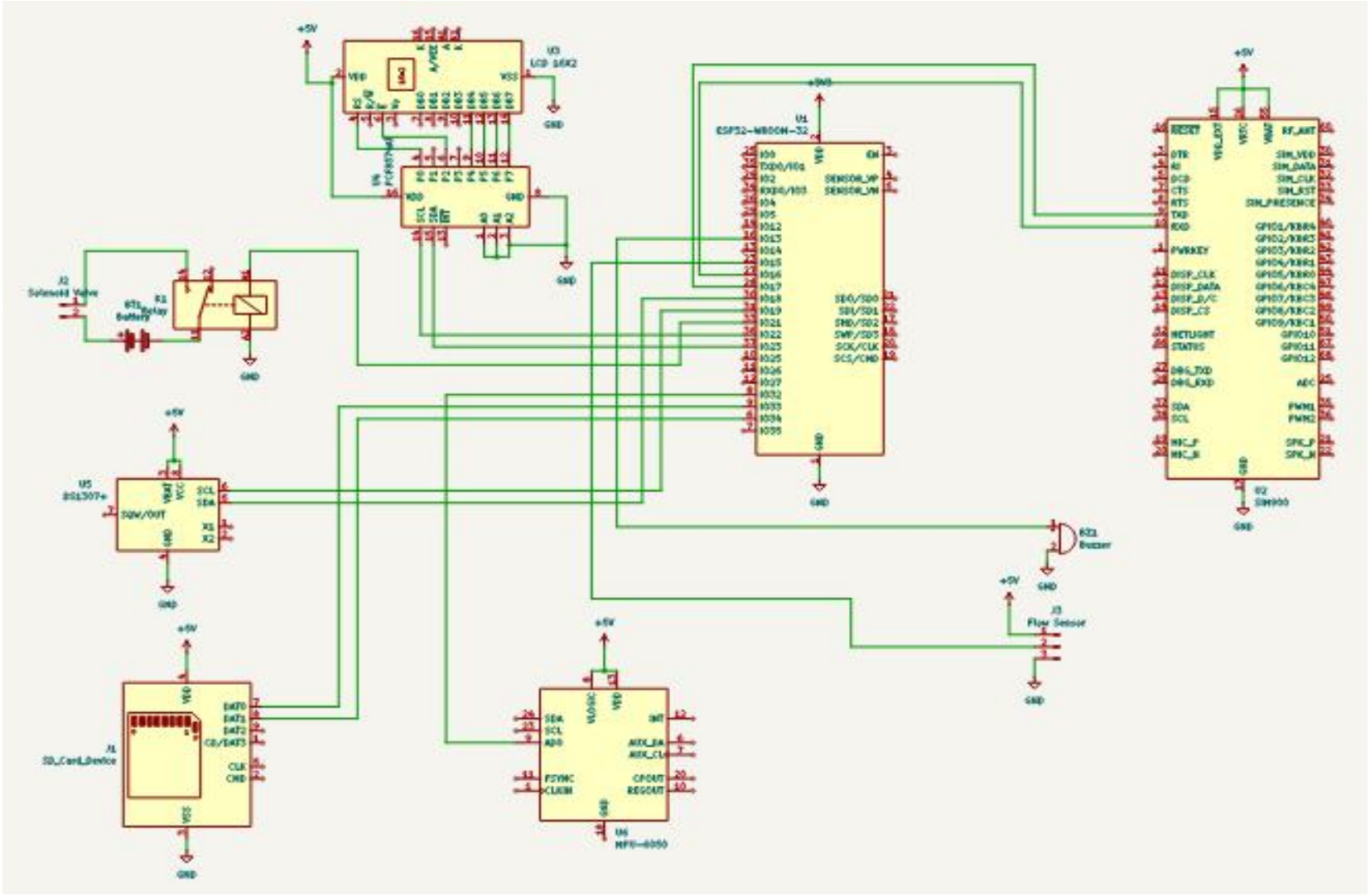


Figure 3.12 Schematic Diagram of the System Prototype

3.6 Circuit Simulation

Figure 3.13 presents the simulated layout of the proposed Smart Water Metering System. The figure consists of an ESP32 microcontroller, a Flow sensor, a Gyroscope, an I2C LCD, a Buzzer, a SIM800L GSM module, a micro-SD card Module, and a 12 V Solenoid valve. The inputs of the circuits are the Flow sensor and the Gyroscope, which are both fed to the microcontroller. Pulses generated by water flow and the opening of the gyroscope, which are detected as tamper, are sent to the ESP32. The microcontroller processes the input data and displays the output on the I2C LCD. The buzzer and SIM800L GSM are also triggered upon tampering events. Moreover, a micro-SD card module is added to store usage data locally, and a solenoid valve is used to control water flow.

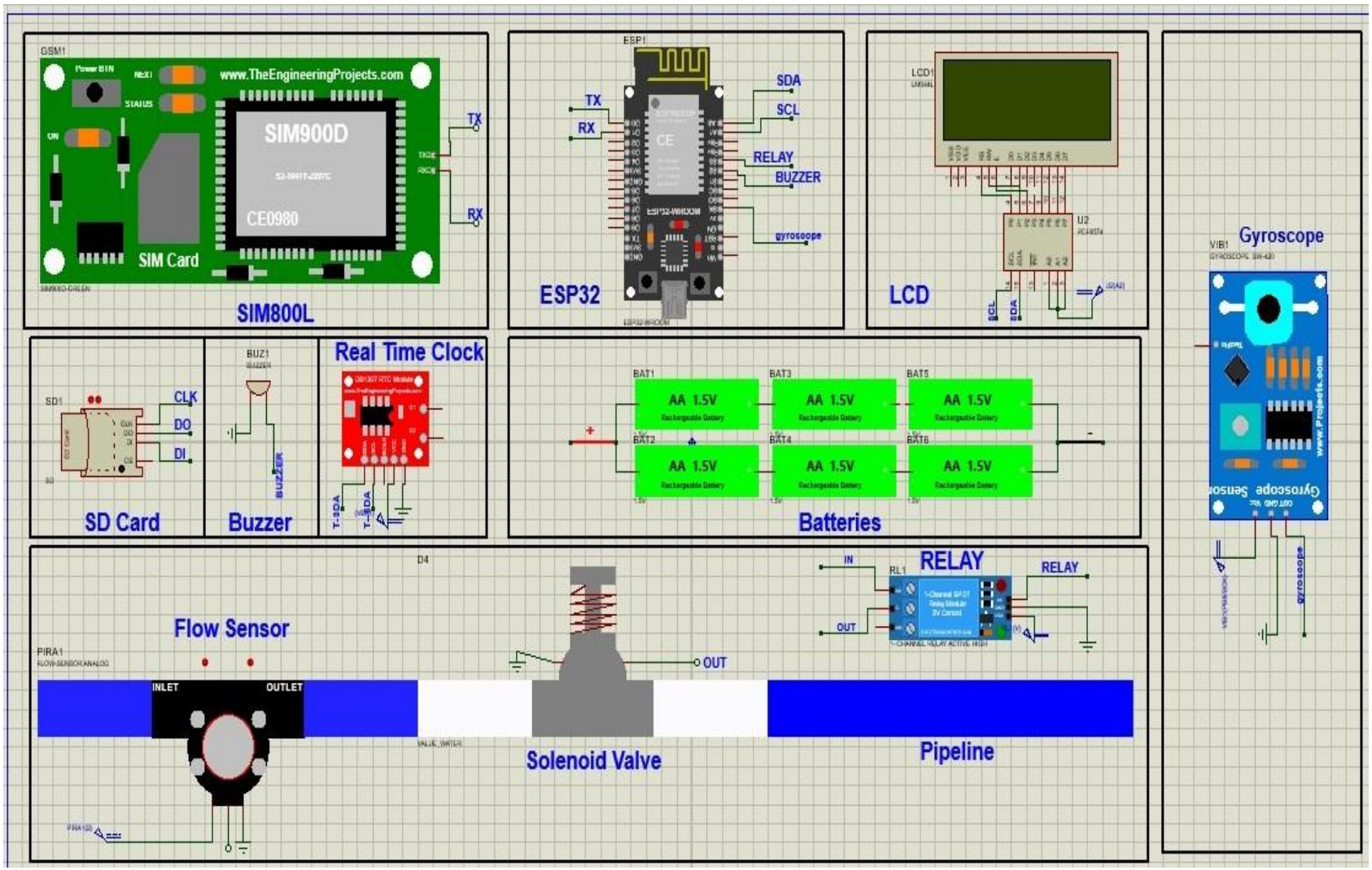


Figure 3.13 The Layout Circuit of the System on Proteus

3.7 Prototype Implementation

A prototype of the smart water metering system is illustrated in Figs. 3.14 and Figure 3.15 The system is composed of a 16x4 I2C LCD, an ESP32 microcontroller with integrated Wi-Fi and Bluetooth capabilities, a YF-S201C flow sensor, a solenoid valve, a relay module, a SIM800L GSM module, a gyroscope sensor, an SD card module, and a buzzer, all interconnected via a breadboard. The flow sensor and gyroscope act as primary inputs to the microcontroller, providing real-time water usage data and detecting physical tampering, respectively. The ESP32 processes these inputs according to embedded control logic, displaying relevant status updates, flow rates, and the gyroscope readings. A relay module interfaces between the microcontroller and the solenoid valve, controlling the valve's operation during tamper events via the Blynk platform. For alerting, the GSM module transmits SMS notifications in the event of tampering, while the buzzer provides an audible signal on-site. The SD card module stores time-stamped usage data and events locally, enhancing system reliability. The cloud interface, accessed via Wi-Fi, uploads data every six hours for analysis and remote control. The breadboard simplifies integration and wiring during the prototyping process. The microcontroller serves as the core processing unit, coordinating the entire operation as per the programmed firmware. The full source code is presented in Appendix I.

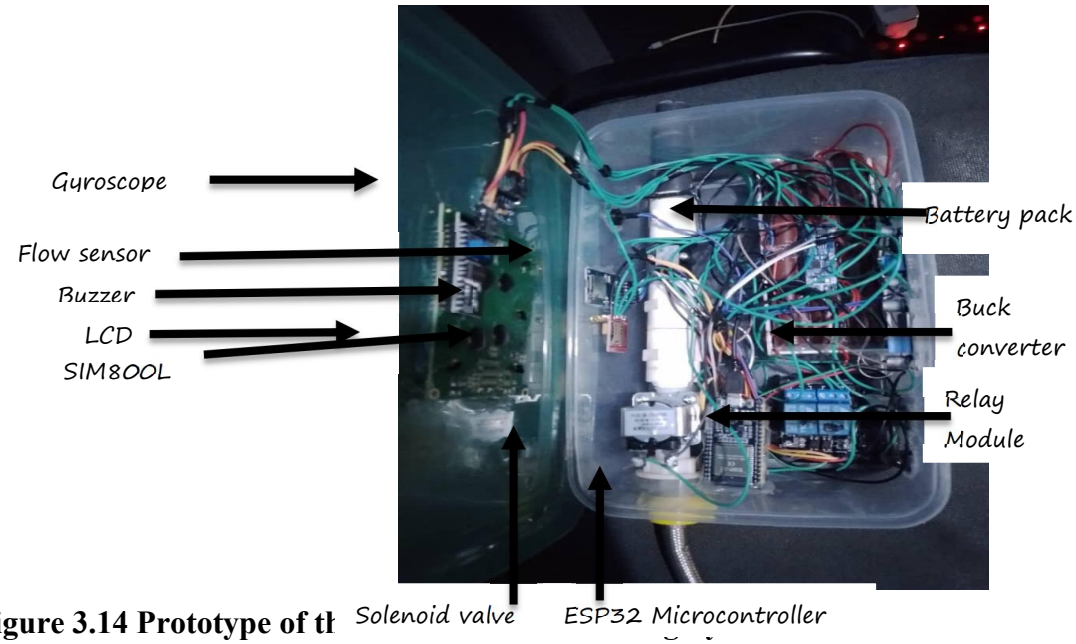


Figure 3.14 Prototype of th

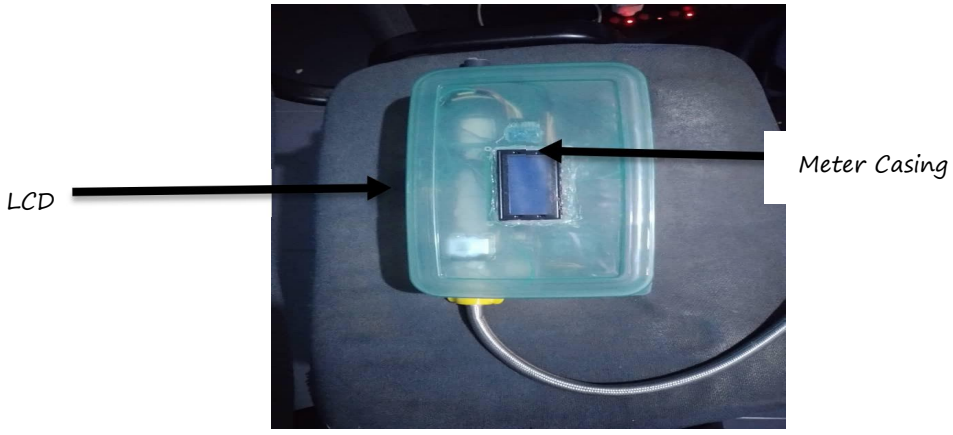


Figure 3.15 Prototype in an Enclosure

3.8 Summary

The smart water metering system was designed and implemented using an ESP32 microcontroller to provide automated water monitoring and control. It integrates a YF-S201 flow sensor for usage readings and a relay-driven solenoid valve for automatic shut-off in the event of leaks. A SIM800L GSM module sends SMS alerts during tampering, while an SD card module logs data locally. An I2C 16x4 LCD provides real-time readings, and a buzzer provides audible alerts. Powered by a 12 V supply stepped down via a buck converter, the system emphasises modular design, energy efficiency, and real-time response to offer a scalable, low-cost water management solution.

4. Results and Discussion

4.1 Overview

This section presents the results obtained during testing and implementation of the proposed smart water metering system under different operating conditions. The system performance is discussed alongside the cost of the implemented prototype.

4.2 Results of System Implementation

The results of the implementation system are categorised in five stages. These include:

System initialisation;
Volume measurement;
Monitoring and control;
Tampering detection; and
Graphical analysis of data.

4.2.1 System Initialisation

Figure 4.1 shows the system's state immediately after power is supplied. The LCD shows the message "Loading System", indicating the initial boot sequence of the ESP32 microcontroller has been successfully initiated. During this phase, the system performs critical startup routines, which include initialising the internal libraries, establishing communication with peripheral components (such as the GSM module, flow sensor, and SD card module), and attempting to connect to a pre-configured Wi-Fi network for cloud communication. Concurrently, the GSM module (SIM800L) initialises and searches for a mobile network to register onto, a necessary step for enabling SMS functionality.

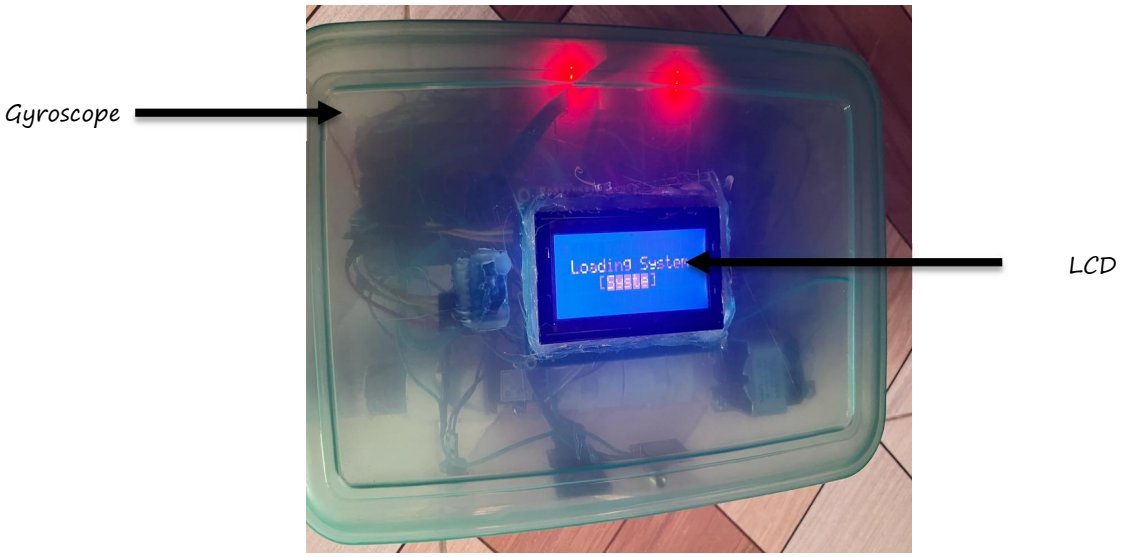


Figure 4.1 System Initialisation



(a) (b)

Figure 4.2 (a) Closed Valve Reading, (b) Opened Valve Reading

4.2.2 Water Volume Measurement

The system can accurately measure water volume and is dependent on the control of the solenoid valve. Figure 4.2 (a) and Figure 4.2 (b) show the LCD readings of total volume when the valve is closed and opened, respectively.

4.2.3 Remote Monitoring and Control

The system successfully controlled the valve to allow water flow remotely via the cloud-based Blynk application dashboard. Upon the operator's command, the Blynk server sends a control signal to the ESP32 microcontroller, instructing it to close and open the valve. Real-time monitoring is achieved through the integration of the Blynk application, as shown in Figure 4.3 and Figure 4.4

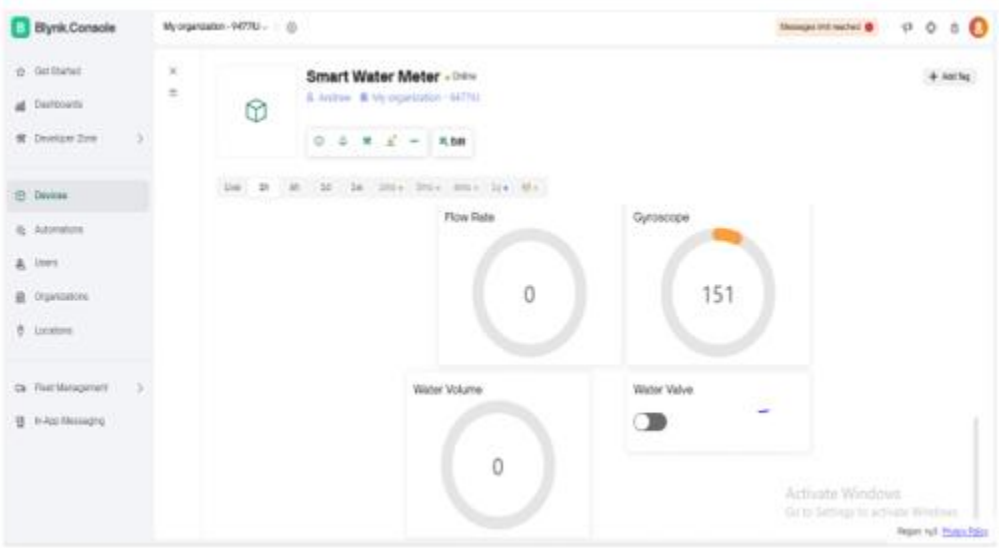


Figure 4.3 Cloud Display After Initialisation

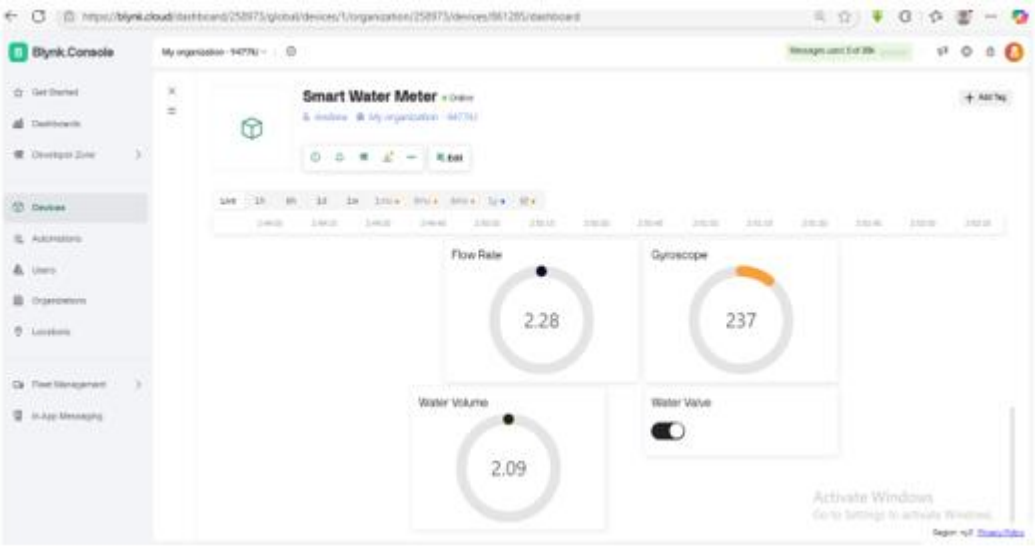


Figure 4.4 Cloud Display When the Valve is Opened

4.2.4 Tampering Detection

To ensure the physical security of the meter and prevent unauthorised movement or manipulation by users, an MPU6050 gyroscope sensor was attached to the meter to detect movement, tilting, and opening of the meter, causing the buzzer to activate and send an alert to the operator when tampering exceeds the threshold value of 300 degree/s. Fi. 4.5 shows an SMS alert after continuous tampering of the meter.



Figure 4.5 Tampering Detection SMS Alert

4.2.5 Graphical Analysis of Data

Graphical display of data in the Blynk application for real-time data analysis was successful when the meter was tested for 2 to 3 hours of water flow. The graph displays hourly water usage analysis, with the X-axis representing hours of the day and the Y-axis indicating the volume of water used. This analysis enables demand forecasting and planning for infrastructure needs, promoting conservation by showing users their water usage habits, as illustrated in Fig 4.6

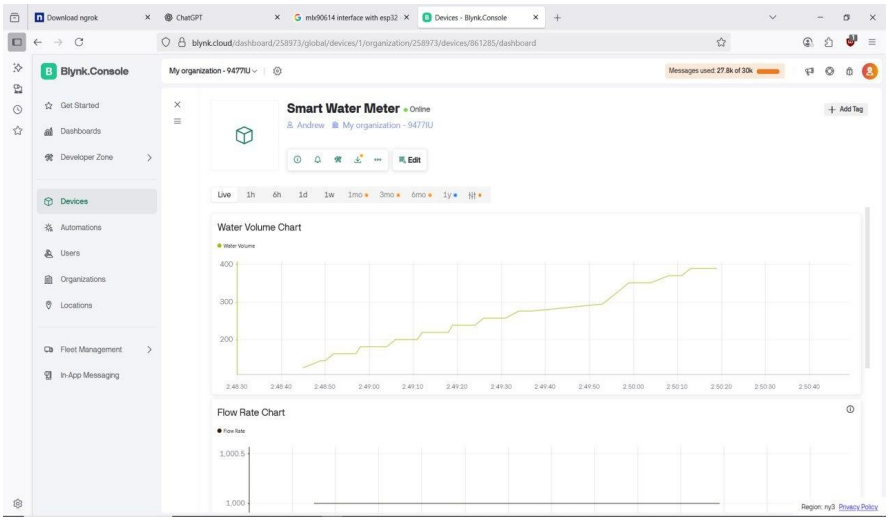


Figure 4.6 Graphical Data Display of Water Consumption

4.3 Cost Analysis

Table 4.1 presents the components used to develop the smart water metering system and their corresponding costs in Ghana cedis (GH₵) and United States dollars (USD). Based on the Oku Electronics online market prices reported in the project, the total component cost was GH₵980.00, equivalent to approximately US$84.12 using the stated exchange rate of GH₵11.65 to US$1.00 on 29 August 2025.

Table 4.1 Summary of the Components Cost Analysis

SN	Component	Qty.	Unit Price (GH₵)	Total (GH₵)	Total (USD)
1.	ESP32 Dev Board	1	165.00	165.00	14.16
2.	SIM800L GSM Module	1	85.00	85.00	7.30
3.	Flow sensor (YF-S201)	1	75.00	75.00	6.44
4.	Gyroscope	1	42.00	42.00	3.61
5.	8-Channel Relay	1	75.00	75.00	6.44
6.	Micro SD card Module	1	22.00	22	1.89
7.	16x4 LCD Screen	1	72.00	72.00	6.18
8.	5 V Active Buzzer	1	8.00	8.00	0.69
9.	Buck converter	4	18.00	72.00	6.18
10.	Rechargeable Lithium Battery	6	49.00	294.00	25.24
11.	Meter Casing	1	25.00	25.00	2.15
12.	Battery Charger Protection Board	1	45.00	45.00	3.86
TOTAL				980.00	**84.12**

Source: Anon. (2025).

4.4 Summary of Findings

Based on the results of implementing the proposed system, it can be stated that the proposed Smart Water Metering System with GSM can:

Accurately measure water consumption in real-time;
Effectively detect and deter tampering attempts through gyroscopic sensing, sounding buzzer, and provide instant SMS alerts;
Enable remote monitoring and control via cloud-based platform (Blynk);
Securely log consumption data both locally (SD card) and in the cloud; and
Be implemented with a cost of GH₵ 980, which is equivalent to USD 84.12.

5. Conclusions, Recommendations and Future Work

5.1 Conclusions

Based on the results of the smart water metering system, the following conclusions can be made:

The meter successfully achieved accurate real-time water flow measurement;
Tampering detection via the gyroscope worked reliably;
The integration of the cloud for monitoring and control was successful; and
Real-time display and data analysis were successful.

5.2 Recommendations

Based on the results presented, it is recommended that:

Utility providers employ this smart water metering to replace traditional meters to improve water management and improve customer satisfaction; and

Smart water Metering System with GSM should be employed to help detect tampering and reduce revenue losses.

5.3 Future Work

Future developments focus on integrating a user interface enabling users to monitor their water consumption for optimal usage, along with machine learning algorithms to predict future trends and assist in detecting leaks.

Acknowledgements

We thank Almighty God for His blessings throughout our academic years. Our deepest gratitude goes to our supervisor, Mr. I. K. M. Prah, for his invaluable guidance and support. We also appreciate the Department of Electrical and Electronic Engineering at UMaT, Essikado, for their resources and encouragement. Special thanks are also due to our family and friends for their unwavering support.

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Appendix A. ESP32 Source Code for the Implementation

Publication note: authentication credentials, Wi-Fi credentials, and the recipient phone number have been replaced with placeholders; the implementation logic is otherwise retained from the source report.

```
#define BLYNK_TEMPLATE_ID "TMPL2w60teFQ0"

#define BLYNK_TEMPLATE_NAME "smart water meter"

#include <SPI.h>

#include <SD.h>

#include <BlynkSimpleEsp32.h>

#include <Wire.h>
```

```
#include <MPU6050.h>
#include <LiquidCrystal_I2C.h>
#include <math.h>
#define SD_CS 5
#define SD_MOSI 23
#define SD_MISO 19
#define SD_SCK 18
#define LCD_ADDRESS 0x27
#define MPU6050_ADDRESS 0x68
#define BUZZER_PIN 12
volatile int pulseCount = 0;
const int flowSensorPin = 14;  // GPIO pin for YF-S201C
const float calibrationFactor = 4.5;
File logFile;
#define NUM_RELAYS 1
int relayPins[NUM_RELAYS] = {27};
const char* deviceNames[NUM_RELAYS] = {
  "Water Valve"};
bool relayStates[NUM_RELAYS] = {false};
// ==== BLYNK CONFIG ====
char auth[] = "YOUR_BLYNK_AUTH_TOKEN";
char ssid[] = "YOUR_WIFI_SSID";
char pass[] = "YOUR_WIFI_PASSWORD";
unsigned long lastTime = 0;
float flowRate = 0.0;
float totalVolume = 0.0;
portMUX_TYPE mux = portMUX_INITIALIZER_UNLOCKED;
MPU6050 mpu;
```

```
LiquidCrystal_I2C lcd(LCD_ADDRESS, 16, 4);
#define MODEM_RX 17  // SIM800 TX -> ESP32 RX (GPIO17)
#define MODEM_TX 16  // SIM800 RX <- ESP32 TX (GPIO16)
#include <HardwareSerial.h>
HardwareSerial sim800(2); // Use UART2
String phoneNumber = "+233XXXXXXXXX"; // Replace with authorised recipient number
BlynkTimer timer;
float filteredGyro = 0;
float alpha = 0.8; // Low-pass filter smoothing
const int NUM_SAMPLES = 30; // For averaging
const float GYRO_THRESHOLD = 300; // Threshold for buzzer alert
int gyroValue;
void showLoadingScreen() {
  lcd.clear();
  lcd.setCursor(0, 1);
  lcd.print(" Loading System ");
  lcd.setCursor(0, 2);
  lcd.print("[     ]");
  for (int i = 0; i < 5; i++) {
    lcd.setCursor(1 + i, 2);
    lcd.write(255); // Full block character
    delay(120);
  }
  delay(500);
}
void showSystemTitle() {
  lcd.clear();
  lcd.setCursor(0, 1);
```

```
  lcd.print(" Water Metering");
  lcd.setCursor(0, 2);
  lcd.print(" System");
  delay(2000);
}
void IRAM_ATTR pulseCounter() {
  portENTER_CRITICAL_ISR(&mux);
  pulseCount++;
  portEXIT_CRITICAL_ISR(&mux);
}
void setup() {
  Wire.begin(21, 22);  // SDA, SCL for ESP32
  Serial.begin(9600);
  Blynk.begin(auth, ssid, pass);
  WiFi.begin(ssid, pass);
  Blynk.config(auth);
  unsigned long startAttemptTime = millis();
  while (WiFi.status() != WL_CONNECTED && millis() - startAttemptTime < 5000) {
  delay(100);
  }
  if (WiFi.status() == WL_CONNECTED) {
  Blynk.connect(1000);  // try to connect for 1 second
  }
  SPI.begin(SD_SCK, SD_MISO, SD_MOSI, SD_CS);  // Use custom SPI pins
  if (!SD.begin(SD_CS)) {
    Serial.println("SD Card initialization failed!");
    lcd.clear();
    lcd.setCursor(0, 0);
```

```
  lcd.print("SD Init Failed");
} else {
  Serial.println("SD Card initialized.");
  lcd.clear();
  lcd.setCursor(0, 0);
  lcd.print("SD Card Ready");
}
timer.setInterval(5000L, logToSDCard); // log every 5 seconds
sim800.begin(9600, SERIAL_8N1, MODEM_RX, MODEM_TX);
pinMode(flowSensorPin, INPUT_PULLUP);
attachInterrupt(digitalPinToInterrupt(flowSensorPin), pulseCounter, RISING);
lcd.init();
lcd.backlight();
for (int i = 0; i < NUM_RELAYS; i++) {
  pinMode(relayPins[i], OUTPUT);
  digitalWrite(relayPins[i], HIGH); // OFF (Active LOW)
}
pinMode(BUZZER_PIN, OUTPUT); // Buzzer pin setup
showLoadingScreen();
showSystemTitle();
mpu.initialize();
if (mpu.testConnection()) {
  lcd.clear();
  lcd.setCursor(0, 0);
  lcd.print("Setup");
  lcd.setCursor(0, 1);
  lcd.print("Completed....");
  Serial.println("MPU6050 connected successfully");
```

```
  } else {
    lcd.clear();
    lcd.setCursor(0, 0);
    lcd.print("Setup Failed");
    Serial.println("MPU6050 connection failed");
    while (1);
  }
  timer.setInterval(3000L, displayGyro);
  timer.setInterval(3000L, displayVolume);
  timer.setInterval(3000L, displayFlow);
  timer.setInterval(1000L, FlowSensor);
  timer.setInterval(2000L, BuzzerTask);
  delay(2000);
  lcd.clear();
}
void loop() {
  if (WiFi.status() == WL_CONNECTED) {
  Blynk.run();
  }
  timer.run();
  gyroValue = getFilteredGyroValue();
  Serial.print("Gyro Magnitude (avg): ");
  Serial.println(gyroValue);
  lcd.clear();
  lcd.setCursor(0, 1);
  lcd.print("  Data Reading:");
  lcd.setCursor(0, 2);
  lcd.print("Gyro: ");
```

```
  lcd.print(gyroValue);
  delay(1000);
  // Clear and Display "Volume"
  lcd.clear();
  lcd.setCursor(0, 1);
  lcd.print("  Data Reading:");
  lcd.setCursor(0, 2);
  lcd.print("Vol: ");
  lcd.print(totalVolume, 2);
  lcd.print(" L");
  delay(1000);
  lcd.clear();
  lcd.setCursor(0, 1);
  lcd.print("  Data Reading:");
  lcd.setCursor(0, 2);
  lcd.print("Flow: ");
  lcd.print(flowRate, 2);
  lcd.print(" L/m");
  delay(1000);
  if (Blynk.connected()) {
  Blynk.virtualWrite(V0, flowRate);
  Blynk.virtualWrite(V3, totalVolume);
  Blynk.virtualWrite(V1, gyroValue);
  }
  Buzzer();
  FlowSensor();
}
// V0 controls Bulb 1 (relay index 1)
```

```
BLYNK_WRITE(V2) {
  int value = param.asInt();
  relayStates[0] = value;
  digitalWrite(relayPins[0], value ? LOW : HIGH);
  Serial.println("Water Valve" + String(value ? "ON" : "OFF"));
}
float getFilteredGyroValue() {
  int16_t gx, gy, gz;
  float sumMagnitude = 0;
  for (int i = 0; i < NUM_SAMPLES; i++) {
    mpu.getRotation(&gx, &gy, &gz);
    float mag = sqrt(gx * gx + gy * gy + gz * gz);
    sumMagnitude += mag;
    delay(20);
  }
  float averageMagnitude = sumMagnitude / NUM_SAMPLES;
  filteredGyro = alpha * filteredGyro + (1 - alpha) * averageMagnitude;
  return filteredGyro;
}
void Buzzer()
{
  // Buzzer alert if above threshold
  if (gyroValue > GYRO_THRESHOLD) {
    digitalWrite(BUZZER_PIN, HIGH);
    delay(500);
    SIM800L();
    delay(500);
  } else {
```

```
    digitalWrite(BUZZER_PIN, LOW);
    delay(500);
  }
  delay(500);
}
void SIM800L() {
  Serial.println("Initializing SIM800L...");
  delay(1000);
  // Basic AT test
  sim800.println("AT");
  delay(1000);
  printResponse();
  // Set SMS to text mode
  sim800.println("AT+CMGF=1");
  delay(1000);
  printResponse();
  // Set recipient
  sim800.print("AT+CMGS=\"");
  sim800.print(phoneNumber);
  sim800.println("\"");
  delay(1000);
  // SMS body
  sim800.print("Alert!!! Meter GWCL-2304187891 is being tempered with.");
  sim800.write(26); // CTRL+Z to send SMS
  Serial.println("SMS sent!");
}
void printResponse() {
  while (sim800.available()) {
```

```
    Serial.write(sim800.read());
  }
}
void FlowSensor()
{
if (millis() - lastTime > 1000) {
    portENTER_CRITICAL(&mux);
    int pulses = pulseCount;
    pulseCount = 0;
    portEXIT_CRITICAL(&mux);
    // Calculate flow rate (L/min)
    flowRate = ((1000.0 / (millis() - lastTime)) * pulses) / calibrationFactor;
    // Flow in Liters/sec
    float flowLitersPerSecond = flowRate / 60.0;
    // Accumulate total volume in Liters
    totalVolume += flowLitersPerSecond;
    lastTime = millis();
    // Serial Monitor Output
    Serial.print("Flow Rate: ");
    Serial.print(flowRate);
    Serial.println(" L/min\t");
    Serial.print("Total Volume: ");
    Serial.print(totalVolume, 3);
    Serial.println(" L");
  }
}
void displayGyro() {
  gyroValue = getFilteredGyroValue();
```

```
  lcd.clear();
  lcd.setCursor(0, 1);
  lcd.print("  Data Reading:");
  lcd.setCursor(0, 2);
  lcd.print("Gyro: ");
  lcd.print(gyroValue);
  if (Blynk.connected()) {
    Blynk.virtualWrite(V1, gyroValue);
  }
}
void displayVolume() {
  lcd.clear();
  lcd.setCursor(0, 1);
  lcd.print("  Data Reading:");
  lcd.setCursor(0, 2);
  lcd.print("Vol: ");
  lcd.print(totalVolume, 2);
  lcd.print(" L");
  if (Blynk.connected()) {
    Blynk.virtualWrite(V3, totalVolume);
  }
}
void displayFlow() {
  lcd.clear();
  lcd.setCursor(0, 1);
  lcd.print("  Data Reading:");
  lcd.setCursor(0, 2);
  lcd.print("Flow: ");
```

```
  lcd.print(flowRate, 2);
  lcd.print(" L/m");
  if (Blynk.connected()) {
    Blynk.virtualWrite(V0, flowRate);
  }}
void BuzzerTask() {
  Buzzer();
}
void logToSDCard() {
  String dataString = "";
  dataString += "Time: ";
  dataString += String(millis() / 1000); // seconds since start
  dataString += "s, Flow: ";
  dataString += String(flowRate, 2);
  dataString += " L/m, Volume: ";
  dataString += String(totalVolume, 2);
  dataString += " L, Gyro: ";
  dataString += String(gyroValue);
  dataString += "\n";
  logFile = SD.open("/log.txt", FILE_APPEND);
  if (logFile) {
    logFile.print(dataString);
    logFile.close();
    Serial.print("Logged to SD: ");
    Serial.print(dataString);
  } else {
    Serial.println("Failed to open log.txt");
  }
```

```
}
```